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SAYA DataX Job Execution Control - User Manual

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Overview

What is Job Execution Control?

Job Execution Control allows you to manage running data ingestion jobs in real-time. You can:

- **Pause** a running job to temporarily halt processing
- **Resume** a paused job to continue from where it stopped
- **Cancel** a job to permanently abandon processing

This feature is essential when you need to:

- Perform scheduled maintenance
- Address system resource constraints
- Stop processing due to data quality issues
- Prioritize urgent jobs over routine ones

Key Benefits

Benefit	Description
 Control	Take immediate action on running jobs without waiting for completion
 Visibility	Track all control actions with a complete audit trail
 Flexibility	Pause and resume jobs seamlessly, preserving progress
 Safety	All actions require a reason, creating accountability
 Efficiency	Stop unnecessary jobs immediately to free resources

Getting Started

Prerequisites

Before using Job Execution Control, ensure you have:

- Access to SAYA DataX platform
- Appropriate permissions:
 - `pause execution` - To pause jobs
 - `resume execution` - To resume jobs
 - `cancel execution` - To cancel jobs

- A running or paused job execution to control

NOTE

Contact your system administrator if you need permission to control job executions.

Accessing the Feature

1. Log in to SAYA DataX platform
2. Navigate to **DataX Workspace**
3. Locate the ingestion job you want to control in the table
4. Click the **More Actions** menu (three vertical dots icon)
5. Select **Control Execution**



Using Job Control Actions

Pausing a Job

Pausing temporarily stops a running job at the next safe checkpoint. The job preserves all progress and can be resumed later.

When to Pause

- Scheduled database maintenance
- System resource constraints (CPU, memory, storage)
- Network bandwidth limitations
- Waiting for dependent jobs to complete
- Testing or troubleshooting scenarios

How to Pause

1. Click **Control Execution** from the job's actions menu
2. In the dialog, click the **Pause** button ( icon with warning color)

NOTE

The Pause button is only enabled when the job status is **IN_PROGRESS**.

3. Enter a reason in the **Reason** field (required)

Example reasons: * "Scheduled maintenance window for database upgrade" * "High CPU usage on server - pausing non-critical jobs" * "Data quality issue detected - pausing for investigation"

4. Review the contextual help message displayed below the reason field
5. Click **Submit**

What Happens

1. The job status changes to **STOPPING**
2. System stops processing at the next chunk boundary
3. Once stopped, the job status changes to **STOPPED**
4. A pause record is added to the Control History

TIP

Chunk Boundary: System processes data in chunks (e.g., 1000 records at a time). When you pause, the job completes its current chunk before stopping, ensuring data consistency.

Expected Timeline

Chunk Size	Typical Delay	Notes
Small (100-1000 records)	1-5 seconds	Quick stop for most jobs
Medium (1000-10000 records)	5-30 seconds	May take longer for complex transformations
Large (10000+ records)	30-60 seconds	Consider using smaller chunk sizes if frequent pauses needed

Resuming a Job

Resuming restarts a paused job from where it stopped, preserving all progress.

When to Resume

- Maintenance window completed
- System resources available again
- Data quality issues resolved
- Ready to continue processing

How to Resume

1. Click **Control Execution** from the job's actions menu

2. In the dialog, click the **Resume** button ( icon with success color)

NOTE

The Resume button is only enabled when the job status is **STOPPED** and was previously paused.

3. Enter a reason in the **Reason** field (required)

Example reasons: * "Maintenance completed - resuming data ingestion" * "System resources stabilized - resuming processing" * "Data quality issue resolved - continuing job"

4. Review the contextual help message

5. Click **Submit**

What Happens

1. System creates a new job execution linked to the original
2. Completed steps are skipped automatically
3. Processing resumes from the last incomplete step
4. The job continues with the same configuration and parameters

IMPORTANT

New Execution ID: Resuming creates a new execution ID. The Control History links both executions together, so you can track the complete job lifecycle.

Expected Behavior

- **Progress Preserved:** All completed chunks are skipped
- **State Maintained:** ExecutionContext (job state) carries over
- **Same Configuration:** Job parameters, readers, writers remain unchanged
- **New Execution:** New job execution ID assigned for tracking

Canceling a Job

Canceling permanently abandons a job. The job cannot be resumed after cancellation.

When to Cancel

- Job launched with incorrect parameters
- Source data is corrupted or invalid
- Business requirement changed - output no longer needed
- Job stuck in error loop
- Duplicate job launched by mistake

CAUTION

CAUTION

Permanent Action: Canceling is irreversible. The job cannot be resumed. If you need to reprocess the data, you must launch a new job.

How to Cancel

1. Click **Control Execution** from the job's actions menu
2. In the dialog, click the **Cancel** button (⊗ icon with error color)

NOTE

The Cancel button is enabled when the job status is **IN_PROGRESS** or **STOPPED**.

3. Enter a reason in the **Reason** field (required)

Example reasons: * "Incorrect source file uploaded - cancelling ingestion" * "Business requirement changed - output no longer needed" * "Job launched with wrong parameters - will relaunch"

4. Review the **warning message** displayed below the reason field
5. Click **Submit**
6. Confirm the action when prompted (if configured)

What Happens

1. The job status immediately changes to **ABANDONED**
2. All processing stops without waiting for chunk completion
3. The job is marked as terminated in the system
4. A cancel record is added to the Control History
5. The job cannot be restarted or resumed

Recovery Options

If you cancel a job by mistake:

1. Launch a **new job** with the same configuration
2. If source files were moved/archived, restore them first
3. Check if any partial data was written and clean up if needed
4. Monitor the new job execution to ensure it processes correctly

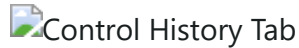
Control History

Viewing Control History

The Control History tab shows all control actions performed on a job execution.

Accessing Control History

1. Navigate to **DataX Workspace**
2. Click on a job execution to open the details panel
3. Select the **Control History** tab



Summary Section

At the top of the Control History tab, you'll see a summary:

- **Activity Icon:** Indicates control actions were taken
- **Total Control Requests:** Count of all control actions
- **Action Breakdown:**
 -  **Pauses:** Number of pause actions (orange indicator)
 -  **Resumes:** Number of resume actions (green indicator)
 -  **Cancel:** Number of cancel actions (red indicator)



Control History Table

The table displays all control actions with the following columns:

Column	Description
Control ID	Unique identifier for the control action
Action	Type of action (PAUSE, RESUME, CANCEL) with color-coded chip
Status	Processing status (REQUESTED, IN_PROGRESS, COMPLETED, FAILED)
Reason	Operator-provided explanation for the action
Created By	Username of the person who initiated the action
Created Date	Timestamp when the action was requested
Failure Message	Error details if the action failed (displayed in red)

Status Indicators

Status	Color	Meaning
REQUESTED	Default	Control action submitted, not yet processed
IN_PROGRESS	Blue	Action is being processed by the system
COMPLETED	Green	Action applied successfully
FAILED	Red	Action failed - see failure message for details

Empty State

If no control actions have been taken on a job:

- The summary shows 0 for all action counts
- The table displays: "No control actions found for this execution"

Understanding Job States

Job Status Flow

```
@startuml
skinparam state {
    BackgroundColor<<running>> LightGreen
    BackgroundColor<<stopped>> LightYellow
    BackgroundColor<<terminal>> LightGray
}

[*] --> IN_PROGRESS : Job Started

IN_PROGRESS <<running>> : Processing data
IN_PROGRESS --> STOPPING : Pause Action

STOPPING <<running>> : Finishing current chunk
STOPPING --> STOPPED : Chunk completed

STOPPED <<stopped>> : Paused - can resume
STOPPED --> IN_PROGRESS : Resume Action\n(new execution)

IN_PROGRESS --> COMPLETED : Normal completion
IN_PROGRESS --> FAILED : Error occurred
IN_PROGRESS --> ABANDONED : Cancel Action

COMPLETED <<terminal>> : Successfully finished
FAILED <<terminal>> : Failed with error
ABANDONED <<terminal>> : Manually cancelled

COMPLETED --> [*]
```



```
FAILED --> [*]  
ABANDONED --> [*]  
  
@endum1
```

Status Definitions

Status	Category	Description
IN_PROGRESS	Running	Job is actively processing data. Can be paused or cancelled.
STOPPING	Running	Pause requested - job finishing current chunk. Automatically transitions to STOPPED.
STOPPED	Paused	Job successfully paused. Can be resumed or cancelled.
COMPLETED	Terminal	Job finished successfully. No actions available.
FAILED	Terminal	Job failed with error. Can be restarted (new job).
ABANDONED	Terminal	Job cancelled by operator. Cannot be restarted.

Action Availability Matrix

Job Status	Pause	Resume	Cancel
IN_PROGRESS	✔ Yes	✗ No	✔ Yes
STOPPING	⌚ In progress	✗ No	✔ Yes
STOPPED	✗ No	✔ Yes	✔ Yes
COMPLETED	✗ No	✗ No	✗ No
FAILED	✗ No	✗ No	✗ No
ABANDONED	✗ No	✗ No	✗ No

Best Practices

When to Use Each Action

Use PAUSE When:

- ✔ You plan to resume the job later

-  The issue is temporary (maintenance, resources)
-  You want to preserve processing progress
-  Multiple jobs need to be paused and resumed together

Use RESUME When:

-  Temporary issue resolved
-  Ready to continue processing
-  System resources available
-  Job was previously paused (not failed)

Use CANCEL When:

-  Job launched with wrong parameters
-  Source data is invalid
-  Output no longer needed
-  Job needs to be restarted from scratch
-  Don't use for temporary issues - use PAUSE instead

Writing Effective Reasons

Good reasons are:

- **Specific:** State the exact issue or requirement
- **Actionable:** Explain what action will be taken
- **Traceable:** Reference ticket numbers or change requests if applicable

Examples

Good Reasons:

- "Scheduled maintenance window DB-2025-12-01 - pausing non-critical jobs"
- "High CPU usage (>90%) detected - pausing to investigate performance issue"
- "Source file corrupted at row 45000 - cancelling and requesting re-upload"
- "INC-12345: Emergency patch deployment requires pause for 30 minutes"

Poor Reasons:

- "pause" (too vague)
- "testing" (no context)

- "stop this" (unclear intent)
- "error" (what error?)

Coordination with Teams

When controlling jobs that affect other teams:

1. **Notify stakeholders** before pausing long-running jobs
2. **Document timeline** in the reason field
3. **Update status** in team communication channels
4. **Track dependencies** to avoid downstream impact
5. **Verify completion** of resumed jobs

Monitoring After Control Actions

After pausing or resuming:

1. **Check Job Status:** Verify the action completed successfully
2. **Review Control History:** Confirm action recorded in audit trail
3. **Monitor Resources:** Ensure system resources normalized
4. **Validate Data:** Check data consistency for resumed jobs
5. **Alert Teams:** Notify dependent processes of status change

Troubleshooting

Common Issues

Issue: Pause Button is Disabled

Symptoms: Cannot click the Pause button in the control dialog.

Possible Causes:

1. Job is not in `IN_PROGRESS` status
2. You lack the `pause` execution permission
3. Another control action is already in progress

Resolution:

- Check job status in the workspace table

- Contact your administrator for permission verification
- Wait for active control action to complete

Issue: Resume Button is Disabled

Symptoms: Cannot click the Resume button in the control dialog.

Possible Causes:

1. Job is not in STOPPED status
2. Job was not previously paused (e.g., it failed naturally)
3. You lack the resume execution permission
4. Another control action is already in progress

Resolution:

- Verify job status is STOPPED
- Check Control History - last action should be successful PAUSE
- If job failed naturally, use "Restart" instead of Resume
- Contact administrator for permission issues

Issue: Job Not Stopping After Pause

Symptoms: Job remains in STOPPING status for extended period.

Possible Causes:

1. Very large chunk size - job finishing current chunk
2. Complex transformation logic causing delay
3. Network latency for remote data sources
4. System under heavy load

Resolution:

- Wait for current chunk to complete (check chunk size in job configuration)
- If stuck for >5 minutes, contact support with job execution ID
- Check system resources (CPU, memory, network)
- Review application logs for errors

Issue: Resume Creates New Execution with Same Status

Symptoms: After resuming, the new execution shows same status as before.

Possible Causes:

1. This is normal behavior - System creates new execution
2. Job may be starting from the same step

Resolution:

- This is expected - new execution ID is assigned
- Check Control History to see link between original and resumed execution
- Monitor step progress to verify job continues from checkpoint

Issue: Control Action Failed

Symptoms: Control History shows FAILED status with error message.

Common Failure Messages:

Error Message	Resolution
"Active control action already exists"	Another control action in progress - wait for it to complete
"Execution must be running"	Job status changed before pause applied - check current status
"Execution not stopped"	Job must be STOPPED to resume - verify current status
"Execution not paused"	Resume only works for paused jobs - use Restart for failed jobs
"Failed to stop job: <details>"	System error - contact support with Control ID

Resolution:

1. Note the Control ID and failure message
2. Check current job status in workspace
3. If persistent, contact support with:
 - Job Execution ID
 - Control ID
 - Failure message
 - Screenshot of Control History

Getting Help

If you encounter issues not covered in this guide:

1. **Check Logs:** Review application logs for error details

2. **Consult Operations Guide:** See [Operations Debugging Guide](#)

3. **Contact Support:** Provide:

- Job Execution ID
- Control ID (if available)
- Action attempted (Pause/Resume/Cancel)
- Error message from Control History
- Screenshots of workspace and control dialog

FAQ

General Questions

Q: Can I pause a job that's almost complete?

A: Yes, but consider if it's worth it. The job will finish its current chunk before stopping. If less than one chunk remains, it may complete before pausing.

Q: What happens to processed data when I pause?

A: All data processed up to the current chunk is committed to the database. When you resume, the job continues from the next unprocessed chunk.

Q: Can I change job parameters when resuming?

A: No, resume uses the same parameters as the original job. If you need different parameters, cancel the job and launch a new one.

Q: How long are control records kept?

A: Control History is retained according to your organization's data retention policy. Typically 90-180 days. Contact your administrator for specific retention settings.

Q: Can I pause multiple jobs at once?

A: Currently, you must pause each job individually. Bulk control operations are planned for a future release.

Q: What's the difference between Cancel and Stop?

A: In SAYA DataX:

- **Pause** = Temporary stop (can resume)
- **Cancel** = Permanent stop (cannot resume)
- There is no separate "Stop" action

Technical Questions

Q: What is a chunk boundary?

A: A chunk is a batch of records processed together (e.g., 1000 records). A chunk boundary is the point between chunks where the job can safely pause. The chunk size is configured per job.

Q: Why does Resume create a new execution ID?

A: System architecture requires a new execution ID for each restart. The Control History links the original and resumed executions for tracking.

Q: Can I resume a job that failed?

A: Resume is only for paused jobs. For failed jobs, use the **Restart** action (separate feature) to restart from the last successful checkpoint.

Q: Is there a timeout for paused jobs?

A: Paused jobs remain stopped indefinitely. However, your organization may have policies requiring jobs to be resumed or cancelled within a specific timeframe.

Q: Can I pause a job in a specific step?

A: Pause operates at the job level, not step level. The job stops at the next chunk boundary within the currently executing step.

Appendix

Keyboard Shortcuts

When the Control Execution dialog is open:

Shortcut	Action
	Navigate between action buttons and reason field
	Submit (when reason field has focus)

Shortcut	Action
	Close dialog without submitting

Glossary

Term	Definition
Job Execution	Single attempt to run a data ingestion job
Chunk	Batch of records processed together in System (e.g., 1000 records)
Checkpoint	Point where job progress is saved, allowing resume from that position
Execution Context	Persistent storage of job state across pause/resume
Control Action	Operator-initiated command (pause/resume/cancel) on a job execution
Audit Trail	Complete history of control actions with reasons and timestamps
Terminal Status	Final job status (COMPLETED, FAILED, ABANDONED) - no further actions possible

Related Documentation

- [Technical Design Document](#)
- [Operations Debugging Guide](#)
- SAYA DataX User Guide - Job Management
- SAYA Security Guide - Permissions and Roles

Revision History

Version	Date	Changes	Author
1.0	December 2025	Initial release	3Cortex Technologies

End of User Manual



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